Machine Learning Experiment Report: Predicting CustomerID

Date: July 18, 2026

Introduction:

Our team recently conducted an automated machine learning experiment to develop a predictive model for the CustomerID target column. The goal of this project was to investigate the feasibility of using machine learning algorithms to predict CustomerID.

Executive Summary:

Our experiment resulted in the development of a Linear Regression model as the best performer. The model's performance is satisfactory, but there is room for improvement. The key findings are:

- **RMSE** (Root Mean Squared Error): 0.87, indicating a relatively high error rate
- **R2** (Coefficient of Determination): 0.045, suggesting that the model only accounts for a small portion of the variance in CustomerID

Model Performance:

Our top-performing Linear Regression model achieved the following metrics:

- RMSE: 0.8709624133280072
- R2: 0.0455742848417916

The features used to train the model were ['Age', 'Annual_Income_k', 'Spending_Score', 'Years_as_Customer', 'Store_Location', 'Purchased_Premium'].

Next Steps:

To improve the model's performance, we recommend the following actions:

1. **Feature Engineering:** Investigate additional features that may improve the model's explanatory power and ability to predict CustomerID.

2. **Model Selection:** Consider alternative regression models, such as Decision Trees and Random Forests, to identify if they perform better than Linear Regression.

3. **Hyperparameter Tuning:** Perform hyperparameter tuning to optimize the model's performance and reduce overfitting.

4. **Data Quality Improvement:** Verify the accuracy and completeness of the data to ensure it is representative of the real-world situation.

By implementing these recommendations, we aim to improve the model's performance and provide valuable insights that can inform business decisions.

Recommendation:

Based on the results of this experiment, we recommend continuing to explore automated machine learning approaches to predict CustomerID. With further refinement and optimization, we are confident that our model can provide actionable insights that drive business growth.

Signature: [Your Name]

Data Scientist